

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/0 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? _____ 129 _____

Does R2 show up as an OSPF neighbor on R1? _Yes_

Does R2 show up as an OSPF neighbor on R3? _No_

What does this information tell you?

All traffic from R3 destined for the 192.168.2.0/24 network is routed via R1. This occurs because R2's S0/0/1 interface is configured as passive, preventing OSPF advertisements from being sent across that link. The total path cost of 129 comes from two T1 serial links (each with a default OSPF cost of 64) plus the GigabitEthernet 0/0 link on R2 (cost 1): $64 + 64 + 1 = 129$.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____ S0/0/1 _____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

65 (One T1 (1.544 Mb/s) serial link (at a cost of 64) plus the R2 Gigabit 0/0 LAN link (at a cost of 1)).

Is R2 listed as an OSPF neighbor to R3? Yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- g. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Modern network equipment supports link speeds well above 100 Mb/s. To ensure more precise OSPF cost calculations for these higher-speed links, you should configure a higher reference-bandwidth value (using the auto-cost reference-bandwidth command).

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

The cost to 192.168.3.0/24: R1 S0/0/1 + R3 G0/0 ($781+1=782$). The cost to 192.168.23.0/30: R1 S0/0/1 and R3 S0/0/1 ($781+64=845$).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1,562. Each serial link now costs 781, and the route to the 192.168.23.0/24 network traverses two serial links. $781 + 781 = 1,562$.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

OSPF will choose the route with the least accumulated cost. The route with the lowest accumulated cost is: R1-S0/0/0 + R2-S0/0/1 + R3-G0/0, or $781 + 781 + 1 = 1,563$. This metric is smaller than the accumulated cost of R1-S0/0/1 + R3-G0/0, or $1565 + 1 = 1,566$.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Controlling the router ID is important because it determines the Designated Router (DR) and Backup Designated Router (BDR) on multi-access networks. If the router ID is automatically selected from a physical interface IP address, it can change if that interface goes down, potentially causing instability, DR/BDR reelection, and temporary routing disruptions. To prevent this, the router ID should be manually set using a loopback interface IP address (which is always up) or with the router-id command for stability and predictability. _____

2. Why is the DR/BDR election process not a concern in this lab?

The DR/BDR election only occurs on multi-access networks (such as Ethernet LANs or Frame Relay) where multiple routers share the same broadcast domain. In this lab, all OSPF connections use serial point-to-point links, which do not require or perform DR/BDR elections—each link forms a simple adjacency with no designated router needed.

3. Why would you want to set an OSPF interface to passive?

Setting an OSPF interface to passive is useful on LAN interfaces connected to end devices (not other routers). It prevents the router from sending or processing OSPF Hello packets on that interface, thereby eliminating unnecessary control traffic, reducing CPU and bandwidth usage, and improving security by blocking potential rogue neighbor attempts. The connected subnet is still advertised to other OSPF neighbors so routing information remains intact.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF's use of areas (especially the requirement for a backbone area 0) makes the protocol hierarchical. By dividing a large network into multiple areas, OSPF reduces the size of the link-state database, limits the scope of link-state advertisements (LSAs), minimizes SPF calculations, and summarizes routing information at area boundaries. This hierarchical design allows OSPF to scale efficiently to very large networks while maintaining fast convergence and low overhead.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0. This single network statement enables OSPF on all interfaces whose IP addresses fall within the 10.0.0.0/8 range (wildcard mask 0.255.255.255 matches any address from 10.0.0.0 to 10.255.255.255) and places them into area 0.

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

~~The RIP route (metric 4) will be used. When multiple routing protocols advertise the same destination, the router installs the route with the lowest administrative distance (AD), not the lowest metric. Default administrative distances are:~~

RIP: 120

OSPF: 110

EIGRP: 90 (for internal routes) EIGRP has the lowest AD (90), followed by OSPF (110), then RIP (120). Therefore, the EIGRP route is preferred and installed in the routing table, regardless of the much higher composite metric (4039043).